

MIT 18.05 Introduction to Probability and Statistics

Cheat sheet 1

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1 The Core Link: Counting and Probability

The **fundamental principle of probability** in a discrete system is based on **counting**: if an experiment has n **equally probable outcomes**, the **probability** of a specific event occurring is the **ratio** of the number of "desirable" outcomes k to the total number of outcomes n ($P = k/n$).

2 Set Theory: The Language of Outcomes

To calculate probabilities, one must first define the "**cast list**" of an experiment: the **sample space**.

- **Sets and Elements**: A set S is a collection of elements, where $x \in S$ denotes membership.
- **Operations**:
 - **Union** ($A \cup B$): Elements in A OR B .
 - **Intersection** ($A \cap B$): Elements in BOTH A and B .
 - **Complement** (A^c): Elements in the universal set S that are NOT in A .
- **DeMorgan's Laws**: These laws describe the relationship between these operations, stating that $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

3 Advanced Counting Techniques

When **sample spaces** become too large to list manually, specific **counting rules** are applied:

- **Rule of Product**: If action 1 has n ways and action 2 has m ways, there are $n \cdot m$ **ways** to perform both in sequence. This holds even if the second action depends on the first, provided the **number of options remains constant**.
- **Inclusion-Exclusion Principle**: To find the size of a union $|A \cup B|$, we use $|A| + |B| - |A \cap B|$. The subtraction is necessary because the intersection is **double-counted** when adding $|A|$ and $|B|$ individually.
- **Permutations** (nPk): Used when the **order matters** (e.g., gold, silver, and bronze medals). **Formula**: $nPk = \frac{n!}{(n-k)!}$.
- **Combinations** (nCk): Used when **order does not matter** (e.g., picking a subset of cards). **Formula**: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

4 Probability Terminology and Axioms

An **experiment** is a repeatable procedure, the **sample space** (Ω) is the set of all possible outcomes, and an **event** is a subset of Ω .

1. **Rule 1**: $0 \leq P(\omega) \leq 1$. Probabilities are **always between 0 and 1**.
2. **Rule 2**: The sum of probabilities of all possible outcomes must **equal 1**.
3. **Complement Rule**: $P(A^c) = 1 - P(A)$.

5 Conditional Probability and Independence

Conditional probability ($P(A|B)$) measures how the probability of an event changes given **”extra information”**. It is visualized as the **proportion of the area of B** that is taken up by A .

- **Formula:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$.
- **Independence:** Two events are independent if knowing one occurred **does not change** the probability of the other: $P(A|B) = P(A)$ or $P(A \cap B) = P(A) \cdot P(B)$.

6 Bayes’ Theorem and Total Probability

6.1 Law of Total Probability

If a sample space is **partitioned into disjoint events** (e.g., B_1, B_2, B_3), the **total probability** of an event A is the **sum of its parts** across those partitions: $P(A) = \sum P(A|B_i)P(B_i)$. This is often organized using **probability trees**, where probabilities are **multiplied along branches** and **summed at the end**.

6.2 Bayes’ Theorem

This theorem allows us to **”invert” conditional probabilities**: $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$.

6.3 The Base Rate Fallacy

A critical explanatory example is the **screening test for a rare disease**. Even with a **95% accurate test**, if a disease only affects **0.5% of the population** (the **base rate**), a **positive test result** might only mean an **8.3% chance** of actually having the disease. This occurs because the number of healthy people who test **”false positive”** far outweighs the small number of sick people who test **”true positive”**.